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Communication Efficient Federated Learning With Quantization-Aware Training Design

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ABSTRACT Model quantization is an effective method that can improve communication efficiency in federated learning (FL). The existing FL quantization protocols almost stay at the level of post-training quantization (PTQ), which comes at the cost of large quantization loss, especially in the setting of low-bits quantization. In this work, we propose a FL quantization training strategy to reduce the impact of quantization on model quality. Specifically, we first apply quantization-aware training (QAT) to FL (QAT-FL), which reduces quantization distortion by adding a fake-quantization module to the model so that the model could perceive future quantization during training. The convergence guarantee of the QAT-FL algorithm is established under certain assumptions. On the basis of the QAT-FL algorithm, we extend the discussion of non-uniform quantization and the adaptive algorithm, so that the model can adaptively adjust the parametric distribution and the number of quantization bits to reduce the amount of traffic in training. Experimental results based on MNIST, CIFAR-10 and FEMNIST datasets show that QAT-FL has advantages in terms of training loss and model inference accuracy, and adaptive-bits quantization of QAT-FL also greatly improves communication efficiency.

INDEX TERMS Federated learning, quantization-aware learning, non-uniform quantization, post-training quantization, adaptive-bits quantization.

I. INTRODUCTION

WITH the explosive growth of computational devices and the rapid development of large models, the scale of networks, and the volume of training data are expanding at an unimaginable speed [1]. Coupled with concerns about the privacy protection of transmitting personal information, the appeal of storing data locally and pushing network computing to the edge is increasing. As a distributed prototype of machine learning, federated learning (FL) has shown great advantages in the training models collaboratively using local resources without data sharing among multiple devices. In recent years, FL has demonstrated significant practical value in privacy protection [2], natural language processing [3], healthcare [4], and other fields.

In normal FL frameworks, nodes consist of a central server and other edge nodes, achieving model exchange through local updates and federated aggregation. Edge nodes optimize models based on local datasets using stochastic gradient descent (SGD) and periodically exchange parameters with the central server to merge local models. The earliest architecture for FL, aimed at preventing user privacy leakage and efficient communication, was proposed in [5]. A major challenge for FL is the communication bottleneck caused by limited bandwidth between nodes due to large-scale training [6].

In order to improve communication efficiency, several methods have been studied. One method for communication efficiency is lossy compression of the gradients or models. The work in [7] proposed and studied the adaptive federated dropout method to reduce the communication costs associated with FL. An efficient FL framework named FedMP was proposed in [8], which can improve both computation and communication efficiency over heterogeneous workers through adaptive model pruning. The work in [9] proposed structured updates and sketched updates to reduce uplink communication costs.

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In recent years, as one of the methods of model compression, model quantization has become a research hotspot to improve communication efficiency. Reference [10] proposed UVeQFed, which utilized universal vector quantization methods to mitigate the effect of limited communication in FL. The work in [11] proposed a uniform quantization method called QSGD, which quantized the parameters of the trained model according to a uniform distribution to achieve unbiasedness. Reference [12] considered the non-uniform distribution of model parameters and studied non-uniform quantization. LM-DFL proposed in [13] set the best quantizer for each node's parameter distribution and adaptively adjusted the quantization level to minimize quantization distortion, and also designed the adaptability of the number of quantization layers. And there was even 1bitSGD schemes that look like extreme compression [14], which only retained the sign information of gradient. Additionally, the work in [15] has conducted some research on adaptive gradient quantization in FL.

To address quantization errors, numerous studies have designed various schemes to mitigate their impact. Focusing on the trade-off between the quantization accuracy of the post-training network and the convergence speed of communication rounds, the work in [16] formulated an optimization problem to achieve a balance between the two. The work in [17] similarly established a multi-objective optimization problem, integrating computational resources, communication resources, and the number of quantization bits, with the aim of adjusting the quantization bits under optimal communication and computational resources to obtain better model performance. Furthermore, [18] introduced compensation mechanisms that use additional memory and communication resources to record errors between clients and the server, and compensated for them at appropriate times. There was also work in [19] that innovatively designed the model aggregation method, combining it with over-the-air computation, to optimize the model performance.

The works outlined above, along with almost all the quantization protocols in FL that existed, remained at the level of post-training quantization (PTQ). All of these applied different quantization methods to the model after local updates were completed. This led to irreversible losses due to quantization, especially when the number of quantization bits was at a low level, the quantization error increased, resulting in slow or even non-convergence of model training, which ultimately led to the reduction of the effectiveness and accuracy of the model [20].

Compared with PTQ, quantization-aware learning (QAT) can improve the inference performance of convolutional neural networks (CNN) after model quantization in centralized learning. A simple QAT framework based on centralized learning was studied in [21]. QAT had very good applications in various fields such as image classification [22] for centralized learning, photonic neural networks [23] and so on. In addition, the work in [24] studied adaptive bit-width QAT which automatically optimized the bit-widths of the weight and activation signals during training for more efficient deep neural network inference. However, it is not known whether the application of QAT to FL can improve compared with the PTQ method. There was little work on the application of QAT in FL. The work in [32]only mentioned the method of using QAT during the construction of FL architecture, but did not provide detailed design and how to integrate it with PTQ. When QAT is used in centralized learning, there is no need to consider the communication or updates of the quantized model. Instead, QAT serves as the final step for the model, performing quantization-aware fine-tuning after multiple rounds of full-precision training. In contrast, the focus of research is on how to design the application of QAT in FL. This involves allowing each round of model communication to use low-precision representations while maintaining good performance. Furthermore, there exist some difficulties in the mathematical discussion of the convergence of QAT in FL.

Based on this observation, we propose a Quantization-Aware Training Federated Learning (QAT-FL) framework, enabling nodes to adapt in advance to future quantization during local training, and train quantization-friendly models. Firstly, we design a QAT-FL framework to reduce quantization distortion at the training level. Then, under the appropriate assumptions, we derive the general convergence bounds of the QAT-FL framework, establish a global convergence guarantee, and show that the convergence bounds of QAT-FL are better than the traditional FL quantization protocol. In order to further improve the communication efficiency, we provide the analysis in terms of communication efficiency and computational efficiency, and we also propose a QAT computational method with non-uniform quantization adapted to parameter distribution and derive the optimal number of quantization layers, and combined with QAT-FL, we propose an adaptive-bits quantization QAT-FL framework. The main contributions of this work are summarized as follows:

- **The FL framework of QAT:** we design a QAT neural network module for communication, which reduces the quantization distortion and improves the accuracy of the model under the given quantization bits. We propose distributing the fine-tuning of QAT across each round, thereby allowing models in each round to communicate using low-precision representations while maintaining good performance.
- **Convergence of QAT-FL:** According to the convergence boundary of FL, the results are generalized to QAT-FL to verify its convergence performance and compare with the traditional quantized FL.
- **Non-uniform of QAT-FL:** A method is designed to optimize the forward quantization and backward propagation gradient computation of non-uniform quantization in QAT-FL that can adapt the parameters. Without heuristic settings or complex neural network learning
- **Proposed adaptive-bits quantization algorithm of QAT-FL:** According to the derived QAT-FL

convergence bound, the analysis is provided in terms of communication efficiency and computational efficiency, and an adaptive quantization bits QAT-FL algorithm is derived and proposed to improve the communication efficiency.

The remainder part of this paper is organized as follows. We introduce the system model of FL in Section II. Section III describes the design of the QAT-FL in detail, and the convergence analysis of QAT-FL, the perspective of computational complexity of QAT-FL and the optimal quantization level of adaptive-bits quantization QAT-FL are discussed in Section IV. Simulations and discussions are given in Section V. The conclusion is given in Section VI.

II. SYSTEM MODEL

In this section, we introduce the traditional centralized federated learning system model and the learning problem, followed by strategies for optimizing the learning problem.

A. LEARNING PROBLEM

Consider a centralized FL (CFL) architecture as shown in Fig. 1. The system consists of N distributed nodes and a central node. We define $\mathcal{N} = \{1, 2, \cdots, N\}$. The N nodes possess distributed datasets $\mathcal{D}_1, \mathcal{D}_2, \cdots, \mathcal{D}_N$. The sizes of the datasets are $D_1, D_2, \cdots, D_N$, respectively, for the dataset of node i is labeled as $\mathcal{D}_i$ with a size of D_i. Without loss of generality, it is assumed that the datasets do not overlap, as $\mathcal{D}_i \cap \mathcal{D}_j = \emptyset (\forall i, j \in \mathcal{N}, i \neq j)$. Therefore, the entire dataset and the size of the dataset can be defined as $\mathcal{D} = \cup \{\mathcal{D}_i\}_{i=1}^{N}$ and $D = \sum_{i=1}^{N} D_i$.

We define $\left(\mathbf{x}_j, y_j\right)$ to represent the j-th sample on dataset D_i.In this context, $\mathbf{x}_j$ represents the input of the sample, and y_j represents the corresponding label type of that sample. In the traditional FL system, methods such as Federated Averaging (FedAvg), the local loss function $F_i(\cdot)$ and the global loss function $F(\cdot)$ are defined as follows:

$$F_i(\mathbf{w}) = \frac{1}{D_i} \sum_{j \in \mathcal{D}_i} f\left(\mathbf{w}, \mathbf{x}_j, y_j\right), \tag{1}$$

$$F(\mathbf{w}) = \frac{1}{N} \sum_{i \in \mathcal{N}} F_i(\mathbf{w}), \tag{2}$$

where $f(\cdot)$ represents the loss function for a single sample.

The target of the optimization problem in FL is to minimize the global loss function in order to obtain the optimal parameters as follows:

$$\mathbf{w}^* = \arg minF(\mathbf{w}). \tag{3}$$

Due to the complexity of machine learning models and the original datasets, obtaining a closed-form solution for the optimization problem (3) is extremely difficult. Therefore, gradient-based iterative algorithms are often used to solve this learning problem.

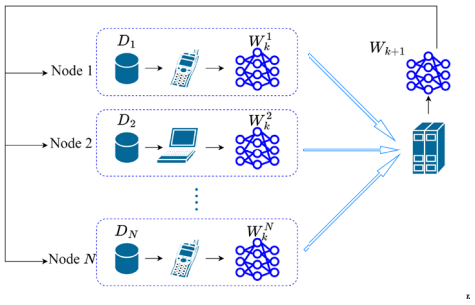


FIGURE 1. An example of FL system model with *N* topology nodes.

B. LEARNING STRATEGY

In CFL, each terminal node downloads the same machine learning model, and its learning strategy can be composed of two steps: local updates and global aggregation. Between each two global aggregations, local updates are executed τ epochs, where τ is referred to as the global aggregation frequency, or the update interval. During the local updates period, each distributed node utilizes its local data to establish a neural network and uses the loss function and SGD to update the parameters. The process can be described as follows:

1) **Local updates:** Each node trains the model on its local dataset using the loss function to compute the gradients. The nodes perform parameter updates using SGD or a variant thereof. The update rule for SGD is typically

$$\widetilde{\mathbf{w}}_{k-1,t}^{(i)} = \widetilde{\mathbf{w}}_{k-1,t-1}^{(i)} - \eta \nabla F_i\left(\widetilde{\mathbf{w}}_{k-1,t-1}\right), \tag{4}$$

where $\widetilde{\mathbf{w}}_{k-1,t}^{(i)}$ represents the local model parameters obtained by the i-th node after the t-th local update round before the k-th round of global model aggregation.

2) **Global Aggregation:** After τ local update rounds, each node sends its updated model parameters $\widetilde{\mathbf{w}}_{k-1,t}^{(i)}$ to the central server. The central server then aggregates these parameters to update the global model as follows:

$$\mathbf{w}_{k,0} = \frac{\sum_{i=1}^{N} D_i \widetilde{\mathbf{w}}_{k-1,\tau}^{(i)}}{D}. \tag{5}$$

And it is then distributed to each decentralized node, which updates its local model parameters accordingly as follows:

$$\widetilde{\mathbf{w}}_{k,0}^{(i)} = \mathbf{w}_{k,0}, i = 1, 2, \cdots, N. \tag{6}$$

3) **Repeat:** The process of local updates and global aggregation are repeated for multiple rounds until the model converges or a stopping criterion is met.

Through modeling analysis, it can be found that FL requires a large number of frequent communication of model parameters, which will bring about the problem of communication efficiency reduction when the communication bandwidth is limited. Therefore, it is necessary to design corresponding strategies, such as quantifying the network parameters.

III. QAT-FL DESIGN

In this section, we propose the QAT-FL design. We will first design the QAT neural network quantization strategy. Then we will design QAT-FL to solve the learning problem expressed in (3) to reduce quantization distortion and improve the model inference capabilities of the FL network.

A. INTRODUCTION OF QUANTIZERS AND DEQUANTIZERS

Considering the floating-point vector $\mathbf{v} \in R^d$ to be quantized, it is quantized into a fixed-point vector, and the vector quantizer $Q(\mathbf{v})$ for the floating-point vector $\mathbf{v}$ after quantization [26]. Taking uniform quantization as an example, the fixed-point and floating-point numbers before and after random quantization and dequantization are commonly related as follows:

$$\mathbf{q}_i = \begin{cases} \left\lfloor \frac{\mathbf{v}_i}{s} \right\rfloor + z, & \text{with probability } 1 + \left\lfloor \frac{\mathbf{v}_i}{s} \right\rfloor - \frac{\mathbf{v}_i}{s} \\ \left\lfloor \frac{\mathbf{v}_i}{s} \right\rfloor + z + 1, & \text{with probability } \frac{\mathbf{v}_i}{s} - \left\lfloor \frac{\mathbf{v}_i}{s} \right\rfloor, \end{cases} \tag{7}$$

$$\mathbf{v}_i' = \left(\mathbf{q}_i - z\right) * s, \tag{8}$$

where $\mathbf{v}$ is the floating-point vector to be quantized, $\mathbf{q}$ is the fixed-point vector after quantization, s is the numerical value of the scaling factor, which determines the step size of the quantization levels. And z is the numerical value of the equivalent zero point, which is the offset from the true zero. (7) is the random quantization process, and (8) is the dequantization process.

For specific quantization methods, s and z have different ways of being solved. Taking QSGD [11] as an example. QSGD is an uniform linear quantizer. The scaling factor and the equivalent zero point are related to the specific composition of the input vector $\mathbf{v}$ and the specific method of solution is as follows:

$$s = \frac{\mathbf{v}_{\max} - \mathbf{v}_{\min}}{\mathbf{q}_{\max} - \mathbf{q}_{\min}}, \tag{9}$$

$$z = \mathbf{q}_{\max} - \mathit{round}\left(\mathbf{v}_{\max}/s\right), \tag{10}$$

where $\mathbf{v}_{\max}$,$\mathbf{v}_{\min}$ are the maximum value and minimum value of the elements in the input vector $\mathbf{v}$, $\mathbf{q}_{\max}$, $\mathbf{q}_{\min}$ are the maximum fixed-value and minimum fixed-value of the elements in the input vector $\mathbf{q}$. Take 8-bit quantization an example, where $\mathbf{q}_{\max} = 127$, $\mathbf{q}_{\min} = -128$ (symmetric) or $\mathbf{q}_{\max} = 255$, $\mathbf{q}_{\min} = 0$ (non-symmetric).

We use C_s to denote the total number of bits for encoding $Q(\mathbf{w})$ and use B to denote the total number of bits that have been communicated by a local client to the central node. Obviously, the scalar in (9) should be encoded with full precision with 32 bits and one bit for the sign of w_i. And the quantizer $Q(\mathbf{w})$ also needs $d\left\lceil \log_2 s \right\rceil$ bits to represent. Therefor, C_s can be given by

$$C_s = d\left\lceil \log_2 s \right\rceil + d + 32. \tag{11}$$

where d is the scale of the network parameters. The purpose of this work is to achieve better model inference performance under the same C_s.

B. ADVANTAGE OF QAT

PTQ and QAT are two main strategies used in the quantization of deep learning models.

PTQ is a process where a pre-trained model is quantized after the completion of training. The model's weights and activations are converted from floating-point to lower-precision representations, typically integers. This is also the common quantization method for most learning scenarios and existing quantization protocols at the level of PTQ in FL, where it can chose different quantizer like QSGD [11], ALQ [12], Lloyd-Max [13].

Compare to PTQ, which quantizes the neural network parameters after training is complete, QAT often results in better model inference. QAT inserts fake-quantization nodes [26] within the regular neural network. Specifically, it adds quantization and dequantization operations after layers in the neural network that require quantization, such as convolutional layers and fully connected layers. This allows the model to learn the quantization that will be used in the future and helps in adjusting the weights to account for the quantization noise.

In order to intuitively understand why QAT can improve the accuracy of quantized models, the work in [27] give a simple example to show that through quantization training, uses fake-quantization nodes and straight-through estimator (STE), which allows the model's weight w to be further updated according to the gradient changes of the quantized weight values with respect to the loss function. This leads the weight to cross some narrow local minima and ultimately converge to a wider local minimum with both smaller loss and wider flatness. As a result, the loss function of the model after quantization becomes smoother relative to the quantized weights. In other words, the neural network learns about the upcoming quantization operation of the model and adapts in advance. The specific analysis about QAT is as follows:

1) **Forward:** Regarding the neural network parameters $\mathbf{w}$ before the fake-quantization nodes, a vector quantizer $Q(\cdot)$ is used. The parameters are quantized to $\mathbf{w}_q$ according to (7) and then dequantized to $\widetilde{\mathbf{w}}$ according to (8). Since the quantization and dequantization functions are not one-to-one correspondences, the neural network parameters of fake-quantized $\widetilde{\mathbf{w}}$ differ from the unquantized parameters $\mathbf{w}$ thereby introducing quantization errors within the neural network.
2) **Backward:** Due to the inability of the quantization process to obtain gradients and propagate gradients (derivatives are almost always zero), we introduce STE [28], [29] to approximate the backward gradient in the quantization function. For uniform quantization, the gradient of the output of the STE direct quantization function is equivalent to the gradient of the input,

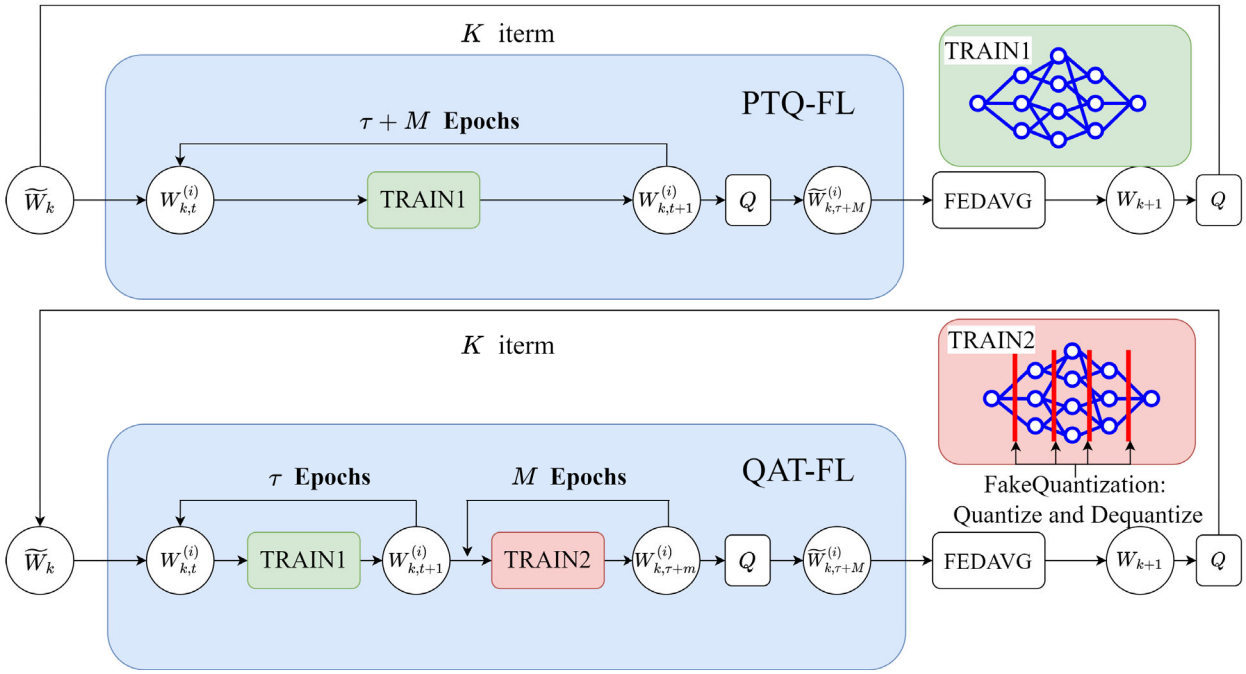


FIGURE 2. QAT-FL system model vs PTQ-FL system model.

that is

$$\frac{\partial L}{\partial \widetilde{\mathbf{w}}_{i,l}} = \frac{\partial L}{\partial \mathbf{w}_{i,l}}, \tag{12}$$

where L is the loss function, $\widetilde{\mathbf{w}}_{i,l}$ and $\mathbf{w}_{i,l}$ denote the quantized and unquantized parameters of the i-th neuron in the l-th layer of the neural network, respectively. Intuitively, STE treats the quantization function as an identity function during the backward pass, allowing the neural network to bypass the quantization layer that cannot be normally processed during the reverse gradient propagation. This approximates the gradient, thereby enabling normal backward gradient propagation to proceed.

Based on the above observations, the integration of the idea of QAT with FL can theoretically achieve the communication efficiency as well as a low error floor of FL.

C. NON-UNIFORM QUANTIZATION AND NON-UNIFORM STE

Due to the non-uniform nature of parameter distribution, sometimes employing uniform quantization methods may lead to suboptimal quantization results. Therefore, data-adaptive non-uniform quantization can be performed. And with the use of non-uniform quantization, the traditional backpropagation calculation method for straight-through estimators cannot be used. According to the derivation of the threshold for the Generalized-STE (G-STE) in Theorem 1 of [30], the forward quantization strategy under non-uniform quantization is as follows:

$$\mathbf{q}_i = \begin{cases} 0, & \mathbf{v}_i < d_0 + a_1/2 \\ j, & d_{j-1} + a_j/2 \le \mathbf{v}_i < d_j + a_{j+1}/2 \\ 2^n - 1, & \mathbf{v}_i \ge d_{2^n-2} + a_{2^n-1}/2, \end{cases} \tag{13}$$

where $d_j = d_0 + \sum_{k=1}^{j+1} a_k, j \in \{1, \ldots, 2^n - 1\}$. During backpropagation, the design of the straight-through estimator, due to the characteristics of non-uniform quantization, requires an additional slope factor to be multiplied on the basis of the original equation relationship, compared to the summation form in (12), that is:

$$\frac{\partial L}{\partial \tilde{w}_{i,l}} = \frac{\partial L}{\partial w_{i,l}} \frac{1}{a_j}, a_{j-1} \le w_{i,l} < a_j. \tag{14}$$

The design of this factor is obtained through heuristic or learning methods in [30], which can reduce the update efficiency of the entire network. By combining our observations and the analysis of (13), we can see that the expressions are highly similar to Lloyd-Max quantization method [13], which is a type of non-uniform quantization. Its specific quantitative expression is as follows:

$$\mathbf{q}_i = \begin{cases} 0, & \mathbf{v}_i < b_1 \\ j, & b_j \le \mathbf{v}_i < b_{j+1} \\ 2^n - 1, & \mathbf{v}_i \ge b_{2^n-1}, \end{cases} \tag{15}$$

where b_j represents the level of reconstruction. And we define l_i to represent the threshold level. The necessary conditions for optimal quantization, derived through analysis, are as follows:

$$b_j = (l_j + l_{j+1})/2, j = 1, 2, \cdots, s - 1, \tag{16}$$

$$l_j = \frac{\int_{b_{j-1}}^{b_j} w\varphi(w)\,dw}{\int_{b_{j-1}}^{b_j} \varphi(w)\,dw}, j = 1, 2, \ldots s, \tag{17}$$

where $\varphi(w)$ is the distribution function of the vector to be quantized. We can see that d_j in (13) is similar to b_j in (16). Similarly, a_j resemble the reconstruction levels l_j in (17). The specific correspondence is as follows:

$$d_{j-1} = b_j - l_{j-1}/2, j = 1, 2, \cdots, s-1, \tag{18}$$

$$a_j = l_{j+1} - l_j, j = 1, 2, \cdots, s-1. \tag{19}$$

For a set of vectors, there are various fast iterative methods to calculate the threshold levels and reconstruction levels in Lloyd-Max. Therefore, we can incorporate this idea into the design of G-STE to avoid more complex parameter learning. For the solution of Lloyd-Max, there are various methods, one of the commonly used methods being the iterative solution based on K-means clustering. Additionally, to ensure consistency, distributed devices also adopt the Lloyd-Max-based PTQ method during communication rounds, so that QAT matches the quantization process.

D. QAT-FL

Based on QAT and vector quantizer, we design QAT-FL to reduce the impact of quantization distortion on FL networks. Each node in QAT-FL performs vector quantizer as discussed in Section III-A and uses the strategy about QAT which we discussed in Section III-B. Thus, one has the follows:

Local update: In the k-th iteration, for each node's local update round $t = 0, 1, \cdots, \tau - 1$. Randomly select r nodes from all S_k nodes to update. At node i, it updates its parameters as follows:

$$\mathbf{w}_{k,t+1}^{(i)} = \mathbf{w}_{k,t}^{(i)} - \eta_{k,t}\widetilde{\nabla} f_i(\mathbf{w}_{k,t}), \tag{20}$$

where $\eta_{k,t}$ is the learning rate. And then, for the QAT round $m = 0, 1, \cdots, M-1$, each node updates its parameters by QAT strategy as follows:

$$\mathbf{w}_{k,\tau+m+1}^{(i)} = \mathbf{w}_{k,\tau+m}^{(i)} - \eta_{k,\tau+m}\widetilde{\nabla} f_i(Q_{fake}(\mathbf{w}_{k,\tau+m})), \tag{21}$$

where the $Q_{fake}(\cdot)$ is the combination of the quantization and dequantization processes we discussed above.

Global aggregation: Nodes send their differential model parameters $Q(w_{k,\tau+M}^{(i)} - w_k)$ by vector quantizer to the central node after τ local update rounds and M QAT rounds. The central node aggregate all of the message with *FedAvg* as follows:

$$\mathbf{w}_{k+1} = \mathbf{w}_k + \frac{1}{r}\sum\nolimits_{i\in S_k} Q_{de}(Q(\mathbf{w}_{k,\tau+M}^{(i)} - \mathbf{w}_k)), \tag{22}$$

where the $Q_{de}(\cdot)$ is the step of dequantization.

We present QAT-FL in Algorithm 1. In addition, we show a comparison of the process frameworks for PTQ-FL and QAT-FL in Fig. 2. Although the introduction of QAT into FL seems straightforward, most of the work using QAT did not give a detailed design and how it can be integrated with PTQ [32], and there are also difficulties in the mathematical discussion of the convergence of QAT-FL.

IV. PERFORMANCE OF QAT-FL

In this section, we will provide the convergence analysis and derivation of QAT-FL. In addition, we also proposed adaptive-bits quantization algorithm QAT-FL to further improve communication efficiency.

A. PRELIMINARIES

Assumption 1: We assume the following conditions (For ease of description, we define the value of $\mathbf{w}$ after quantization and dequantization as $Q(\mathbf{w})$.):

1) $Q(\cdot)$ is unbiased, and its variance increases with the square of the two-norm of its parameters, i.e., $\mathbb{E}[Q(\mathbf{w})|\mathbf{w}] = \mathbf{w}$, $\mathbb{E}\left[\|Q(\mathbf{w}) - \mathbf{w}\|^2|\mathbf{w}\right] \le q\|\mathbf{w}\|^2$.
2) $f_i(w)$ is L-smooth, i.e., $\left\|\nabla f_i(\mathbf{w}) - \nabla f_i(\hat{\mathbf{w}})\right\| \le L\left\|\mathbf{w} - \hat{\mathbf{w}}\right\|$, for some $L > 0$, and any $\mathbf{w}, \mathbf{y}, i$.
3) Gradient estimation is unbiased for stochastic mini-batch sampling, i.e., $\mathbb{E}\left[\widetilde{\nabla} f_i(\mathbf{w})\right] = \nabla f_i(\mathbf{w})$ and the variance of gradient estimation is bound for σ^2, i.e., $\mathbb{E}_\xi\left[\left\|\widetilde{\nabla} f_i(\mathbf{w}) - \nabla f_i(\mathbf{w})\right\|^2\right] \le \sigma^2$.

Definition 1 (Parameters of model): Except the local update round $t = 0, 1, \cdots, \tau - 1$ and global aggregation round $k = 0, 1, \cdots, K-1$, we introduce a new parameter $m = 0, 1, \cdots, M-1$ to refer the QAT round. And we have some definition as follows:

$$\begin{aligned}
\mathbf{w}_{k+1} &= \mathbf{w}_k + \frac{1}{r}\sum_{i\in S_k} Q\left(\mathbf{w}_{k,M+\tau}^{(i)} - \mathbf{w}_k\right),\\
\hat{\mathbf{w}}_{k+1} &= \mathbf{w}_k + \frac{1}{n}\sum_{i\in[n]} Q\left(\mathbf{w}_{k,M+\tau}^{(i)} - \mathbf{w}_k\right),\\
\bar{\mathbf{w}}_{k+1,t} &= \frac{1}{n}\sum_{i\in n}\mathbf{w}_{k,t}^{(i)},\\
\bar{\mathbf{w}}_{k+1,\tau+m} &= \frac{1}{n}\sum_{i\in n}\mathbf{w}_{k,\tau+m}^{(i)}.
\end{aligned} \tag{23}$$

In networks such as CNN, due to the complexity of the network, the loss function is not necessarily non-convex, and SGD may converges to the local minima point, and the expectation of the average value of the gradient norm [31] is often used as an indicator of convergence in research.

B. CONVERGENCE OF QAT-FL

Theorem 1 (Convergence of QAT-FL): Consider the average models over iteration K according to the QAT-FL method outlined in Algorithm 1 Suppose the conditions 1-3 in Assumption 1 are satisfied with i.i.d data distribution, then the expectation of the gradient norm average after K iterations is bounded as

$$\begin{aligned}
&\frac{1}{K(\tau+M)}\sum_{k=0}^{K}\left(\sum_{t=0}^{\tau-1}\mathbb{E}\left\|\nabla f\left(\bar{\mathbf{w}}_{k,t}\right)\right\|^2\right.\\
&\left.+\sum_{m=0}^{M-1}\mathbb{E}\left\|\nabla f\left(\bar{\mathbf{w}}_{k,\tau+m}\right)\right\|^2\right)\\
&\le \frac{2(f(\mathbf{w}_0) - f^*)}{\eta K(\tau+M)} + \eta L(1+q)\left(\frac{n-r}{r(n-1)}4\sigma^2 + \frac{\sigma^2}{n}\right)
\end{aligned}$$

Algorithm 1 QAT-FL

Input

Total number of iterations K, learning rate η, the number of local update round τ and the number of local QAT round M, the number of nodes selected per iteration r.

for $k = 0, 1, \cdots, K-1$ **do**
  Randomly select r nodes S_k from all nodes for updates.
  Central node transmits $\mathbf{w}_k$ to the selected nodes S_k.
  for $i \in S_k$ **do**
    $\mathbf{w}_{k,0}^{(i)} = \mathbf{w}_k$
    for $t = 0, 1, \cdots, \tau - 1$ **do**
      Calculating stochastic gradient descent for conventional neural networks without inserting fake-quantization nodes.
      $\mathbf{w}_{k,t+1}^{(i)} = \mathbf{w}_{k,t}^{(i)} - \eta_{k,t} \widetilde{\nabla} f_i(\mathbf{w}_{k,t})$
    end for
    for $m = 0, 1, \cdots, M-1$ **do**
      Calculating stochastic gradient descent with inserting Fake-quantization nodes.
      $\mathbf{w}_{k,\tau+m+1}^{(i)} = \mathbf{w}_{k,\tau+m}^{(i)} - \eta_{k,\tau+m} \widetilde{\nabla} f_i(Q_{fake}(\mathbf{w}_{k,\tau+m}))$
    end for
    Quantizes the model parameters by quantizer to get $Q(\mathbf{w}_{k,\tau+M}^{(i)} - \mathbf{w}_k)$ and then send it to the central node.
  end for
  FedAvg
  $\mathbf{w}_{k+1} = \mathbf{w}_k + \frac{1}{r} \sum_{i \in S_k} Q_{de}(Q(\mathbf{w}_{k,\tau+M}^{(i)} - \mathbf{w}_k))$
end for

$$+ \eta^2 \frac{\sigma^2}{n}(n+1)(\tau + M - 1)L^2. \tag{24}$$

Proof: The proof of Theorem 1 is deferred to the Appendix.

Remark 1 (The Discussion about the Extension to the Non-IID Data): Extending to the non-i.i.d scenario is not trivial because when different clients perform local updates, the directions of the locally estimated gradients become diversified, and thus the aggregated gradient descent direction may not be the correct one. The discussion on non-IID data is further considered in the work of [32], which highlights a critical challenge in Clustered Federated Learning with Differential Privacy (DP). This work showed that privacy constraints and cluster FL frameworks also significantly affect communication efficiency and convergence. By replacing the weak assumption that the gradient variance is upper bounded with a stricter assumption that the gradient norm is bounded, a convergence result can be derived. Relaxing the uniform variance bound $\mathbb{E}\left[\widetilde{\nabla} f_i(\mathbf{w})\right] = \nabla f_i(\mathbf{w})$ and the variance of gradient estimation is bound for σ^2, i.e., $\mathbb{E}_\xi \left[\left\|\widetilde{\nabla} f_i(\mathbf{w}) - \nabla f_i(\mathbf{w})\right\|^2\right] \leq \sigma^2$ to different variance bounds σ_i^2 in condition 3, our bound can be easily extended to this case and get the following result

$$\begin{aligned}
&\frac{1}{K(\tau+M)} \sum_{k=0}^{K} \left(\sum_{t=0}^{\tau-1} \mathbb{E}\left\|\nabla f\left(\bar{\mathbf{w}}_{k,t}\right)\right\|^2 \right. \\
&\left. + \sum_{m=0}^{M-1} \mathbb{E}\left\|\nabla f\left(\bar{\mathbf{w}}_{k,\tau+m}\right)\right\|^2 \right) \\
&\leq \frac{2(f(\mathbf{w}_0) - f^*)}{\eta K(\tau+M)} + \eta L(1+q)\left(\frac{n-r}{r(n-1)} 4\sigma_a^2 + \frac{\sigma_a^2}{n}\right) \\
&\quad + \eta^2 \frac{\sigma_a^2}{n}(n+1)(\tau+M-1)L^2.
\end{aligned} \tag{25}$$

where $\sigma_a^2 = \frac{1}{n} \sum_{i=0}^{n} \sigma_i^2$. Additionally, we will perform experiments on non-IID data like the FEMNIST dataset, which inherently has data heterogeneity, for experimental comparison in our experiments. And the primary objective of this paper is to demonstrate that our proposed QAT-FL method outperforms the PTQ method. Both methods can be compared horizontally under the same i.i.d setting.

Remark 2 (Advantages of QAT-FL in convergence): Consider the PTQ-FL with the $\tau+M$ local update epochs, then the expectation of the gradient norm average after K iterations is bounded as

$$\begin{aligned}
&\frac{1}{K(\tau+M)} \sum_{k=0}^{K} \left(\sum_{t=0}^{\tau+M-1} \mathbb{E}\left\|\nabla f\left(\bar{\mathbf{w}}_{k,t}\right)\right\|^2 \right) \\
&\leq \frac{2(f(x_0) - f^*)}{\eta K(\tau+M)} + \eta L(1+q)\left(\frac{n-r}{r(n-1)} 4\sigma^2 + \frac{\sigma^2}{n}\right) \\
&\quad + \eta^2 \frac{\sigma^2}{n}(n+1)(\tau+M-1)L^2.
\end{aligned} \tag{26}$$

When comparing the upper bounds of PTQ-FL and QAT-FL, although they appear to be the same on the surface, this is because, during the convergence proof, in order to achieve a more elegant conclusion, the terms $\left\|\nabla f_i\left(\mathbf{w}_{k,t}^{(i)}\right)\right\|^2$ and $\left\|\nabla f_i\left(Q(\mathbf{w}_{k,t}^{(i)})\right)\right\|^2$ were relaxed in (56) in Appendix. This step clearly highlights the fundamental difference between PTQ and QAT: QAT uses the quantized values of the model as the basis for gradient descent calculations. This causes parameters that were originally in the low-loss region of the loss function to change from small gradients to large gradients due to quantization, thereby lowering the upper bound compared to PTQ. This also demonstrates that our proof clearly confirms the principle of QAT convergence.

Modeling the computational complexity of neural networks and federated learning is quite complex, as they are closely related to specific quantization methods, neural network model design, network architecture, and hardware devices. In terms of model training, a common approach is to use floating-point operations (FLOPs) for calculation. For example, for a convolutional layer, if the input image size is C_{in} and the output size is (C_{out}, H, W), the corresponding FLOPs can be calculated as $2C_{in}C_{out}K^2 \times H \times W$, where K is the size of the convolutional kernel. For a fully

connected layer, if the input size is n_{in} and the output size is n_{out}, the corresponding FLOPs are $2n_{in}n_{out}$. Specifically, taking the CNN network of FEMNIST as an example, its FLOPs are 12.3M and the number of parameters is 1.69M. From a formula perspective, quantization does not increase or decrease FLOPs, but it can significantly reduce the model's inference time. For example, for FEMNIST inference on a 12th Gen Intel(R) Core(TM) i7-12700K, the inference time for traditional PTQ is approximately 1.499ms, with the convolutional layers accounting for a significant portion, taking about 800us. In contrast, our QAT-FL, with 4-bit quantization, only requires about 605us for inference, with the convolutional layers taking only 170us, greatly improving the model's inference efficiency.

C. ADAPTIVE-BITS QUANTIZATION QAT-FL

Although QAT-FL can achieve better convergence than PTQ-FL at the same communication bits, the network convergence is very slow if low-bit quantization is used in the communication process. And if the high-quantization bit is used for quantization, it will converge faster, but it will require more communication budget. In any case, the quantization of fixed bits will lead to a decrease in communication efficiency.

In order to improve the communication efficiency of QAT-FL, we propose an adaptive QAT-FL to adapt the number of quantization levels s so that it can adapt the quantization bits n_b during the training.

Corollary 1 (Convergence of QAT-FL based on communicated bits): Based on the above definition, we can get follows:

$$\begin{aligned}&\frac{C_s}{B(\tau+M)}\sum_{k=0}^{(B/C_s)}\left(\sum_{t=0}^{\tau+M-1}\mathbb{E}\left\|\nabla f\left(\tilde{\mathbf{w}}_{k,t}\right)\right\|^2\right)\\&\leq\frac{2\left(f\left(\mathbf{w}_0\right)-f^*\right)C_s}{\eta B(\tau+M)}+\eta L\left(1+q_s\right)\left(\frac{n-r}{r(n-1)}4\sigma^2+\frac{\sigma^2}{n}\right)\\&\quad+\eta^2\frac{\sigma^2}{n}(n+1)\tau(\tau-1)L^2\frac{1}{M+\tau}\\&\quad+\eta^2\frac{\sigma^2}{n}(n+1)M(M-1)L^2\frac{1}{M+\tau},\end{aligned}\tag{27}$$

where $q_s = d/s^2$ in QSGD.

Proof: The result can be derived from Lemma 1 straightforwardly.

Corollary 2 (The adaptive quantization levels QAT-FL strategy): The adaptive quantization levels QAT-FL strategy is gived as follows:

$$s_k^* = \sqrt{\frac{f(\mathbf{w}_0)}{f(\mathbf{w}_k)}}s_0.\tag{28}$$

Proof: We obtain the optimal s to minimize the convergence bound above by differentiating the right-hand side of (27) with respect to s, we have

$$s^* = \sqrt{\frac{A_1}{A_2[f(\mathbf{w}_0)-f^*]}},\tag{29}$$

where $A_1 = \eta^2L\left(4n\left(n-r\right)+r\left(n-1\right)\right)\sigma^2B\left(\tau+M\right)\times\log_e(2)$ and $A_2 = rn\left(n-1\right)$. For k-th, the $\mathbf{w}_k$ can be regarded as a new initialization $\mathbf{w}_0$, and each node communicates B_0 bits with central node in each interval, so the optimal number of quantization levels of global iteration k is

$$s_k^* = \sqrt{\frac{A_1^0}{A_2[f(\mathbf{w}_k)-f^*]}},\tag{30}$$

where $A_1^0 = \eta^2L\left(4n\left(n-r\right)+r\left(n-1\right)\right)\sigma^2B_0\left(\tau+M\right)\times\log_e(2)$. And with the assumption that $f^* = 0$, we can divide s_k^* by s_0^*, the proof of Corollary 2 is complete.

From (28), adaptive-bits quantization QAT-FL employs an ascending number of quantized levels to achieve optimal convergence. And that the adaptive-bits can be given by

$$n_b^k = \left\lceil\log_2 s_k^*\right\rceil.\tag{31}$$

Compared with QAT-FL with fixed number of quantization-level approaches, the adaptive-bits quantization QAT-FL just needs fewer bits for communication to achieve the same convergence performance. But adaptive-bits quantization QAT-FL need to pay additional communication bits to inform the specific quantization bits of the current round.

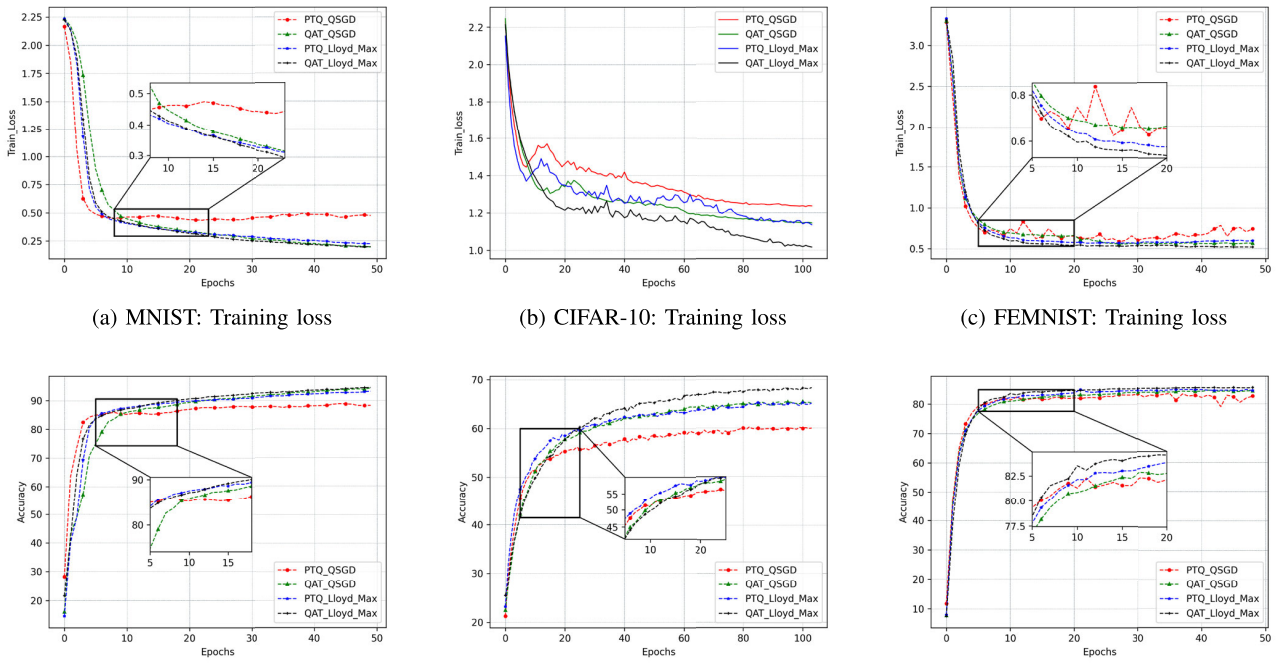
V. SIMULATION AND DISCUSSION

In this section, we present the simulation results of the proposed QAT-FL and adaptive-bits quantization QAT-FL framework.

A. SETUP

In order to evaluate the performance of QAT-FL, we first need to build an FL network and conduct our experiments using the built centralized network. Using python environment and pytorch platform, we establish an FL framework with 100 nodes. All 100 nodes have local datasets. Based on the local datasets, we randomly choose some of the local nodes to train their models and after τ local update rounds and M QAT rounds, each node communicates its differential model parameter with the global node. Also note that during the setup, all of the local nodes have the same structure for model aggregating and the training parameters. For example, QAT-FL sets the same learning rate η and the same number of quantization levels.

1) Baselines: Regarding the selection and analysis of baselines, we have conducted research and implemented various methods such as SignSGD and FedPAQ. We have focused on analyzing the SignSGD method SignSGD: It is a method that achieves network training by only transmitting the sign of each batch training gradient. Each gradient update transmits either 1 (indicating a gradient greater than 0) or -1 (indicating a gradient less than 0). The global node then updates by adding or subtracting a predetermined learning rate. It is evident that this method is highly dependent on the learning rate. If the learning rate is set too small or too large, the model training will encounter problems. Moreover, the design



(a) MNIST: Training loss (b) CIFAR-10: Training loss (c) FEMNIST: Training loss

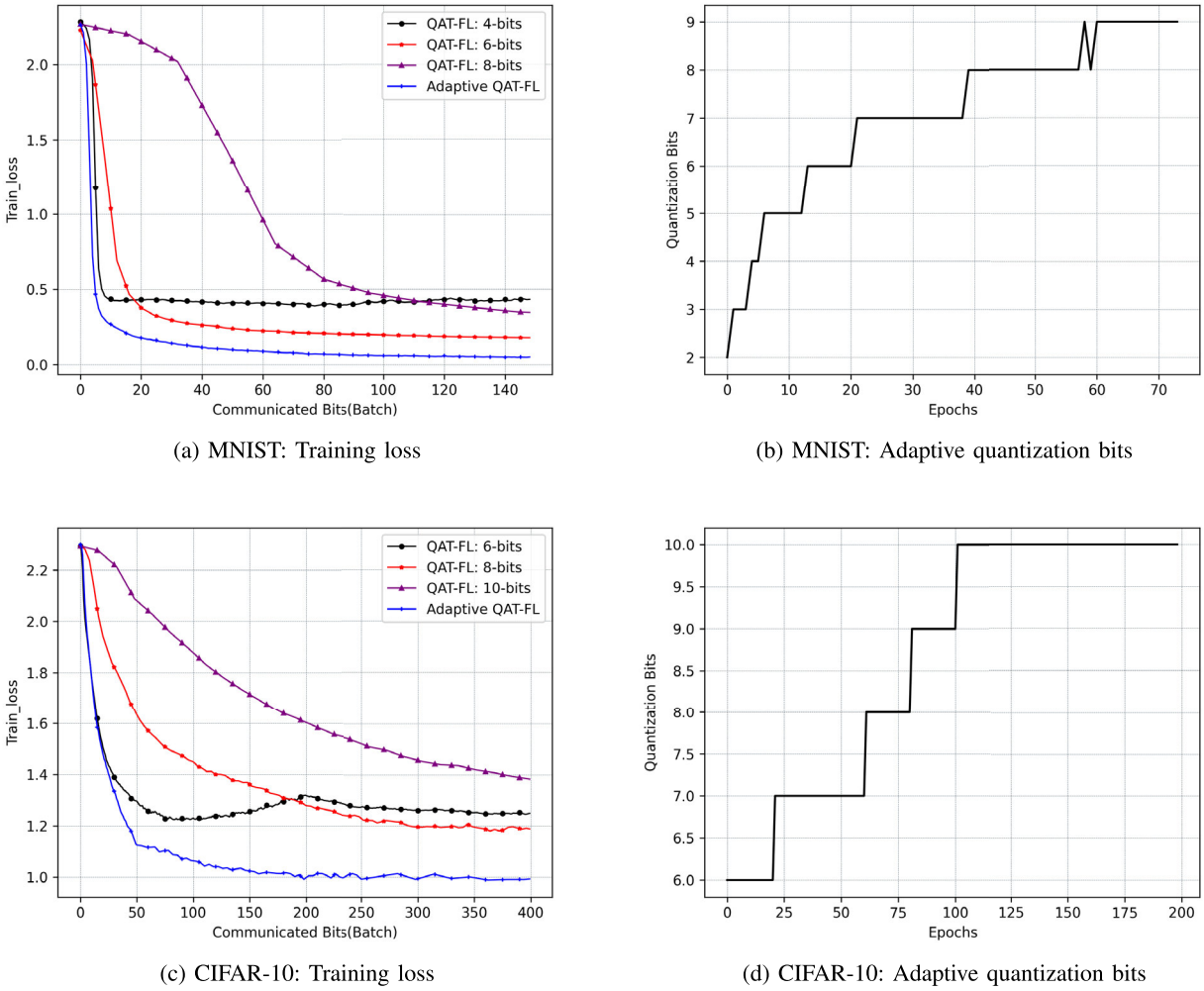
(d) MNIST: Accuracy (e) CIFAR-10: Accuracy (f) FEMNIST: Accuracy

FIGURE 3. Experiments on MNIST, CIFAR-10 and FEMNIST datasets with CNN under different approaches.

of the learning rate varies significantly for different datasets and neural networks. Additionally, the choice of learning rate is closely related to the number of training rounds. Most works and engineering experiments use a stepwise decreasing learning rate for experimentation. We have also conducted various measurements of neural network hyperparameters and learning rates for the SignSGD method. However, we did not obtain satisfactory model inference results. For instance, when we used a stepwise decreasing learning rate for multiple rounds of experiments on FEMNIST, we found that the model's inference accuracy could only reach a maximum of 40%-50%, with significant fluctuations. Therefore, we do not consider SignSGD as a suitable baseline for comparison. And FedPAQ in [33] is essentially the conventional PTQ-FL scheme mentioned in our paper, which employs fixed-bit uniform quantization. Our experimental baseline is also based on this, referring to the non-uniform quantization PTQ scheme in [13] for experimentation. This method is already one of the best in terms of convergence among current PTQ-FL schemes. Therefore, we believe there are no issues with the selection of the baseline scheme. So we compare the QAT-FL with the following baseline approaches: PTQ-FL: We deploy the model just by training it with $\tau + M$ local update rounds and communicate its model parameters, which are quantized by using QSGD and non-uniform quantization with the global node. In the comparison, the number of quantization levels is fixed. We use PTQ-FL with the QSGD quantizer and non-uniform quantizer under different quantization levels as the baselines for QAT-FL.

2) Model and Datasets: We evaluate two different model based on two different datasets. We deploy two different CNN for model training. The two models are trained, tested, and verified by using MNIST and CIFAR-10 dataset. Considering the non-i.i.d. characteristics of real data, we have used experiments and discussions using the FEMNIST dataset. This dataset is partitioned based on EMNIST and can be divided in either an i.i.d. or non-i.i.d. manner. The non-i.i.d. partitioning is based on different writers and includes handwritten characters from female volunteers with diverse backgrounds and occupations. This allows for a more realistic non-i.i.d. distribution. It comprises 62 classes (10 digits, 26 uppercase letters, and 26 lowercase letters) with a total of 805,263 samples, making it more complex and further demonstrating the richness and authenticity of our experiments.

3) Training and Quantization Hyperparameters: In the experiments, we set the local update learning rate $\eta_1 = 0.002$ on MNIST with the QAT learning rate $\eta_2 = 0.002$, and $\eta_1 = 0.01$ on CIFAR-10 and FEMNIST with the QAT learning rate $\eta_2 = 0.01$. And the number of quantization bits $n_b = 3, 4, 5$ on MNIST with 2-bits for the adaptive quantization initial bit experiment, and $n_b = 6, 7, 8$ on CIFAR-10 with 6-bits for the adaptive quantization initial bit experiment. We choose different number of local updates and the number of QAT rounds but all totally $\tau + M = 4$ on MNIST and $\tau + M = 6$ on CIFAR-10 and set the initial value of model parameter with an initialization of Gaussian distribution. We use mini-batch SGD in local updates synchronously.



(a) MNIST: Training loss

(b) MNIST: Adaptive quantization bits

(c) CIFAR-10: Training loss

(d) CIFAR-10: Adaptive quantization bits

FIGURE 4. Experiments on adaptive quantization bits QAT based on MNIST and CIFAR-10 datasets.

B. EVALUATION

In this section, we first simulate QAT-FL and its baselines and the results can verify that the QAT-FL effectively improves the reduced estimation accuracy of training networks caused by communication quantization distortion. And then we design simulation of adaptive quantization bits QAT-FL and compare it with QAT-FL with the fixed bits.

1) Results of QAT-FL: The training loss curves iteration are presented in Fig. 3(a) and Fig. 3(b) based on MNIST and CIFAR-10. The training loss of all different local update rounds sets on QAT-FL converge gradually to a minimal value than PTQ-FL with iteration.

Table 1 shows the accuracy of the model based on MNIST and CIFAR-10 datasets after multiple training epochs with different QAT-FL settings and various quantization bits. We can find that QAT-FL can improve about 2%-9% and 2%-4% of the testing accuracy based on MNIST and CIFAR-10.

From Fig. 3(d) and Fig. 3(e) which show the testing accuracy of the models with the specific number of quantization bits (4-bit on MNIST and 8-bit on CIFAR-10) of training epochs of QAT-FL and PTQ-FL. We can find that during the early epochs of training, the estimation accuracy of QAT-FL may be lower than that of PTQ-FL. Although the QAT scheme in the early stage of updating requires fewer rounds

TABLE 1. Accuracies(%) for $E = 50$ and $\eta = 0.002$ on MNIST, $E = 150$ and $\eta = 0.01$ on CIFAR10.

MNIST	n_b=3(%)	n_{bits}=4(%)	n_b=5 (%)
PTQ-FL($\tau + M = 4$)	80.47	89.75	93.20
QAT-FL($\tau = 0, M = 4$)	84.39	92.38	94.02
QAT-FL($\tau = 1, M = 3$)	87.98	92.58	94.26
QAT-FL($\tau = 3, M = 1$)	84.17	92.29	94.13
QAT-FL($\tau = 2, M = 2$)	89.41	92.80	95.09
CIFAR10	n_b=6 (%)	n_b=7(%)	n_b=8(%)
PTQ-FL($\tau + M = 4$)	59.57	60.76	62.96
QAT-FL($\tau = 0, M = 4$)	61.23	62.97	63.43
QAT-FL($\tau = 1, M = 3$)	63.75	64.56	65.04
QAT-FL($\tau = 3, M = 1$)	64.12	65.78	66.45
QAT-FL($\tau = 2, M = 2$)	63.89	64.24	64.89

for model updating compared to PTQ under the same computational load, it will eventually converge to better model performance, which aligns with our analysis of the rational convergence of QAT.

The results on the complex heterogeneous FEMNIST dataset show that our proposed QAT-FL method exhibits curve changes consistent with the experimental performance and outperforms the standard PTQ method under the same data distribution setting of non-i.i.d.

2) Non-uniform quantization QAT-FL: The result in Fig. 3 show that use more advanced quantization methods, for QAT-FL, under the same number of quantization bits, there will be improvements in model inference accuracy and reductions in loss compared to QSGD. Specifically, non-uniform QAT-FL can achieve approximately 1% increase in inference accuracy compared to non-uniform PTQ-FL, and the training loss is also reduced by 20%-40% on MNIST. However, when comparing to QAT-FL with QSGD, the magnitude of improvement is not substantial. This is because for QAT-FL, as analyzed and implemented earlier, it can be seen that models quantized after QAT and further refined through learning have significantly reduced quantization errors, achieving performance close to that of non-quantized models.

3) Adaptive-bits quantization QAT-FL: In this simulation, we evaluate the training loss and the classification accuracy of adaptive QAT-FL to verify the analysis that the adaptive-bits quantization QAT-FL we designed can converge to a lower loss and a higher accuracy than fixed QAT-FL, and the ascending number of quantization levels can achieve convergence with less bits. We define the network to communicate model volumes of 4-bits or 6-bits as a batch on MNIST or CIFAR-10. Fig. 4 shows the variable quantized bit for a single model based on QAT-FL. We can see that adaptive QAT-FL achieves the best convergence performance under any communicated bits compared with any case of QAT-FL with fixed bits. For example, the training loss of adaptive QAT-FL based on MNIST reduces by almost 50% when the communicated bits is 80 batches. Although low quantization bits can complete multiple rounds of updates with the same amount of communication no matter what, the final convergence level of low-precision quantization will always be lower than that of high-precision quantization. The result in Fig. 4(b) and Fig. 4(d) inspire that QAT-FL with the ascending number of quantization levels can save communicated bits compared with other fixed setting of quantization bits.

VI. CONCLUSION

In this paper, we have proposed QAT-FL algorithm with low quantization distortion. During the training terms for local nodes, QAT-FL inserts a fake-quantization module into the network to adjust the network parameters for the quantization adaptation that will occur in the future. We have derived the quantization distortion and convergence upper bound and the computational complexity of the QAT-FL, which shows its superiority over the existing post-training quantized of FL. And we propose a method to optimize the forward quantization and backward propagation gradient computation of non-uniform quantization in QAT-FL to adapt the parameters. On this basis, we have analyzed and discussed the adaptive algorithm of QAT-FL, which jointly considers quantization-perceptual training and adaptive number of quantization bits in the training process to achieve the training loss and accuracy of the target with fewer communication bits. Finally, on the MNIST, CIFAR-10 and FEMNIST datasets, based on CNN, the experimental results have verified our theoretical analysis of QAT-FL and adaptive-bits quantization of QAT-FL.

APPENDIX

PROOF OF THEOREM 1

Lemma 1: For any global aggregation period $k = 0, 1, \cdots, K-1$, due to the condition 1 in Assumption 1 being valid, we have

$$\mathbb{E}f\left(\mathbf{w}_{k+1}\right) \leq \mathbb{E}f\left(\bar{\mathbf{w}}_{k,\tau+M}\right) + \frac{L}{2}\mathbb{E}\left\|\hat{\mathbf{w}}_{k+1} - \bar{\mathbf{w}}_{k,\tau+M}\right\|^2 + \frac{L}{2}\mathbb{E}\left\|\hat{\mathbf{w}}_{k+1} - \mathbf{w}_{k+1}\right\|^2. \tag{32}$$

Proof: For $f(\cdot)$ which satisfy the condition 2 in Assumption 1, for any $\mathbf{x}$,$\mathbf{y}$

$$f(\mathbf{x}) \leq f(\mathbf{y}) + \langle \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y}\rangle + \frac{L}{2}\|\mathbf{x} - \mathbf{y}\|^2. \tag{33}$$

So we have

$$\begin{aligned} f\left(\mathbf{w}_{k+1}\right) &= f\left(\hat{\mathbf{w}}_{k+1} + \mathbf{w}_{k+1} - \hat{\mathbf{w}}_{k+1}\right) \\ &\leq f\left(\hat{\mathbf{w}}_{k+1}\right) + \left\langle \nabla f\left(\hat{\mathbf{w}}_{k+1}\right), \mathbf{w}_{k+1} - \hat{w}_{k+1}\right\rangle \\ &\quad + \frac{L}{2}\left\|\mathbf{w}_{k+1} - \hat{\mathbf{w}}_{k+1}\right\|^2. \end{aligned} \tag{34}$$

Taking expectations on both sides of the inequality, since $\hat{\mathbf{w}}_{k+1}$ is an unbiased estimator of $\mathbf{w}_{k+1}$, i.e., $\mathbb{E}_{S_k}\mathbf{w}_{k+1} = \hat{\mathbf{w}}_{k+1}$, so follows:

$$\mathbb{E}f\left(\mathbf{w}_{k+1}\right) \leq \mathbb{E}f\left(\hat{\mathbf{w}}_{k+1}\right) + \frac{L}{2}\mathbb{E}\left\|\mathbf{w}_{k+1} - \hat{\mathbf{w}}_{k+1}\right\|^2. \tag{35}$$

According the condition 1 in Assumption 1, $\hat{\mathbf{w}}_{k+1}$ is an unbiased estimator of $\bar{\mathbf{w}}_{k+1,\tau+M}$, i.e.,$\mathbb{E}_Q\hat{\mathbf{w}}_{k+1} = \bar{\mathbf{w}}_{k+1,\tau+M}$, so as follows:

$$\mathbb{E}f\left(\hat{\mathbf{w}}_{k+1}\right) \leq \mathbb{E}f\left(\bar{\mathbf{w}}_{k+1,\tau+M}\right) + \frac{L}{2}\mathbb{E}\left\|\hat{\mathbf{w}}_{k+1} - \bar{\mathbf{w}}_{k+1,\tau+M}\right\|^2. \tag{36}$$

Combining (35) and (36)), Lemma 1 is proven.

Lemma 2: For any global aggregation period $k = 0, 1, \cdots, K-1$, due to the condition 2-3 in Assumption 1 being valid, we have

$$\mathbb{E}f(\bar{\mathbf{w}}_{k,\tau+M}) \leq \mathbb{E}f\left(\mathbf{w}_k\right) - \frac{1}{2}\eta\sum_{t=0}^{\tau-1}\mathbb{E}\left\|\nabla f\left(\bar{\mathbf{w}}_{k,t}\right)\right\|^2 - \frac{1}{2}\eta\sum_{m=0}^{M-1}\left\|\nabla f\left(\bar{\mathbf{w}}_{k,\tau+m}\right)\right\|^2$$

$$
\begin{aligned}
&- \eta\left(\frac{1}{2n} - \frac{1}{2n}L\eta - \frac{1}{n}L^2(M+\tau)(M+\tau-1)\eta^2)\right) \\
&\times \sum_{t=0}^{\tau-1}\sum_{i\in[n]}\mathbb{E}\left\|\nabla f\left(\mathbf{w}_{k,t}^{(i)}\right)\right\|^2 \\
&- \eta\left(\frac{1}{2n} - \frac{1}{2n}L\eta - \frac{1}{n}L^2(M+\tau)(M+\tau-1))\eta^2\right) \\
&\times \sum_{m=0}^{M-1}\sum_{i\in[n]}\mathbb{E}\left\|\nabla f\left(Q\left(\mathbf{w}_{k,\tau+m}^{(i)}\right)\right)\right\|^2 \\
&+ \eta^2\frac{L}{2}\frac{\sigma^2}{n}(M+\tau) \\
&+ \eta^3\frac{\sigma^2}{n}(n+1)(M+\tau)(M+\tau-1)L^2. \qquad (37)
\end{aligned}
$$

Proof: According to the QAT-FL method outlined in Algorithm 1, for any local update period $k = 0, 1, \cdots, K-1$, The equivalent average model is

$$
\bar{\mathbf{w}}_{k,t+1} = \bar{\mathbf{w}}_{k,t} - \eta\frac{1}{n}\sum_{i\in[n]}\tilde{\nabla}f_i\left(\mathbf{w}_{k,t}^{(i)}\right). \qquad (38)
$$

Besides, according to the QAT-FL method outlined in Algorithm 1, for any local QAT round $m = 0, 1, \cdots, M-1$, the equivalent average model is

$$
\bar{\mathbf{w}}_{k,\tau+m+1} = \bar{\mathbf{w}}_{k,\tau+m} - \eta\frac{1}{n}\sum_{i\in[n]}\tilde{\nabla}f_i\left(Q\left(\mathbf{w}_{k,\tau+m}^{(i)}\right)\right). \qquad (39)
$$

As same as Lemma 1, we have

$$
\begin{aligned}
&f\left(\bar{\mathbf{w}}_{k,t+1}\right) \\
\le& f\left(\bar{\mathbf{w}}_{k,t}\right) - \eta\left\langle\nabla f\left(\bar{\mathbf{w}}_{k,t}\right), \frac{1}{n}\sum_{i\in[n]}\tilde{\nabla}f_i\left(\mathbf{w}_{k,t}^{(i)}\right)\right\rangle \\
&+ \eta^2\frac{L}{2}\left\|\frac{1}{n}\sum_{i\in[n]}\tilde{\nabla}f_i\left(\mathbf{w}_{k,t}^{(i)}\right)\right\|^2. \qquad (40) \\
&f\left(\bar{\mathbf{w}}_{k,\tau+m+1}\right) \\
\le& f\left(\bar{\mathbf{w}}_{k,\tau+m}\right) \\
&- \eta\left\langle\nabla f\left(\bar{\mathbf{w}}_{k,\tau+m}\right), \frac{1}{n}\sum_{i\in[n]}\tilde{\nabla}f_i\left(Q\left(\mathbf{w}_{k,\tau+m}^{(i)}\right)\right)\right\rangle \\
&+ \eta^2\frac{L}{2}\left\|\frac{1}{n}\sum_{i\in[n]}\tilde{\nabla}f_i\left(Q\left(\mathbf{w}_{k,\tau+m}^{(i)}\right)\right)\right\|^2. \qquad (41)
\end{aligned}
$$

Using the vector inner product triangle rule $2\langle\mathbf{a},\mathbf{b}\rangle = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - \|\mathbf{a}-\mathbf{b}\|^2$, expectations are taken for the inner product terms

$$
\begin{aligned}
&2\mathbb{E}\left\langle\nabla f\left(\bar{\mathbf{w}}_{k,t}\right), \frac{1}{n}\sum_{i\in[n]}\tilde{\nabla}f_i\left(\mathbf{w}_{k,t}^{(i)}\right)\right\rangle \\
&= \frac{1}{n}\sum_{i\in[n]}2\mathbb{E}\left\langle\nabla f\left(\bar{\mathbf{w}}_{k,t}\right), \nabla f\left(\mathbf{w}_{k,t}^{(i)}\right)\right\rangle \\
&= \mathbb{E}\left\|\nabla f\left(\bar{\mathbf{w}}_{k,t}\right)\right\|^2 + \frac{1}{n}\sum_{i\in[n]}\mathbb{E}\left\|\nabla f\left(\mathbf{w}_{k,t}^{(i)}\right)\right\|^2 \\
&\quad - \frac{1}{n}\sum_{i\in[n]}\mathbb{E}\left\|\nabla f\left(\bar{\mathbf{w}}_{k,t}\right) - \nabla f\left(\mathbf{w}_{k,t}^{(i)}\right)\right\|^2. \qquad (42)
\end{aligned}
$$

And with the help of the smoothness assumption, the third term in (42)

$$
\begin{aligned}
&\left\|\nabla f\left(\bar{\mathbf{w}}_{k,t}\right) - \nabla f\left(\mathbf{w}_{k,t}^{(i)}\right)\right\|^2 \\
&\le L^2\left\|\bar{\mathbf{w}}_{k,t} - \mathbf{w}_{k,t}^{(i)}\right\|^2. \qquad (43)
\end{aligned}
$$

Additionally, according to the definition of local updates, the local model $\mathbf{w}_{k,t}^{(i)}$ and the equivalent average model $\bar{\mathbf{w}}_{k,t}$ are defined as follows:

$$
\mathbf{w}_{k,t}^{(i)} = \begin{cases} \mathbf{w}_{k,0}^{(i)} - \eta\sum_{j=0}^{t}\tilde{\nabla}f_i\left(\mathbf{w}_{k,j}^{(i)}\right), & t \le \tau \\ \mathbf{w}_{k,\tau}^{(i)} - \eta\tilde{\nabla}\sum_{j=\tau}^{t}f_i\left(Q\left(\mathbf{w}_{k,j}^{(i)}\right)\right), & t > \tau \end{cases} \qquad (44)
$$

$$
\bar{\mathbf{w}}_{k,t} = \begin{cases} \mathbf{w}_{k,0} - \eta\sum_{j=0}^{t}\left(\frac{1}{n}\sum_{j\in[n]}\tilde{\nabla}f_j\left(\mathbf{w}_{k,j}^{(i)}\right)\right), & t \le \tau \\ \bar{\mathbf{w}}_{k,\tau} - \eta\sum_{j=\tau}^{t}\left(\frac{1}{n}\sum_{j\in[n]}\tilde{\nabla}f_j\left(Q\left(\mathbf{w}_{k,j}^{(i)}\right)\right)\right), & t > \tau \end{cases} \qquad (45)
$$

Therefore, the expected bias between the local model and the average model when $t \le \tau$ can be expressed as

$$
\begin{aligned}
&\mathbb{E}\left\|\bar{\mathbf{w}}_{k,t} - \mathbf{w}_{k,t}^{(i)}\right\|^2 \\
&\le 2\eta^2\left(\frac{1}{n}\left\|\sum_{j\in[n]}\nabla f\left(\mathbf{w}_{k,0}^{(j)}\right) + \sum_{j\in[n]}\nabla f\left(\mathbf{w}_{k,1}^{(j)}\right)\right.\right. \\
&\quad \left.\left. + \cdots + \sum_{j\in[n]}\nabla f\left(\mathbf{w}_{k,t-1}^{(j)}\right)\right\|^2 + t\frac{\sigma^2}{n}\right) \\
&\quad + 2\eta^2\left(t\sigma^2 + \left\|\nabla f\left(\mathbf{w}_{k,0}^{(i)}\right) + \nabla f\left(\mathbf{w}_{k,1}^{(i)}\right)\right.\right. \\
&\quad \left.\left. + \cdots + \nabla f\left(\mathbf{w}_{k,t-1}^{(i)}\right)\right\|^2\right) \\
&\le 2\eta^2 t\frac{\sigma^2}{n} + 2\eta^2 t\left(\frac{1}{n}\sum_{j\in[n]}\left\|\nabla f\left(\mathbf{w}_{k,0}^{(j)}\right)\right\|^2\right. \\
&\quad + \frac{1}{n}\sum_{j\in[n]}\left\|\nabla f\left(\mathbf{w}_{k,1}^{(j)}\right)\right\|^2 \\
&\quad \left. + \cdots + \frac{1}{n}\sum_{j\in[n]}\left\|\nabla f\left(\mathbf{w}_{k,t-1}^{(j)}\right)\right\|^2\right) + 2\eta^2 t\sigma^2 \\
&\quad + 2\eta^2 t\left(\left\|\nabla f\left(\mathbf{w}_{k,0}^{(i)}\right)\right\|^2 + \left\|\nabla f\left(\mathbf{w}_{k,1}^{(i)}\right)\right\|^2\right.
\end{aligned}
$$

$$+\cdots+\left\|\nabla f\left(\mathbf{w}_{k,t-1}^{(i)}\right)\right\|^2\Bigg). \tag{46}$$

And when $t>\tau$

$$\begin{aligned}
&\mathbb{E}\left\|\bar{\mathbf{w}}_{k,t}-\mathbf{w}_{k,t}^{(i)}\right\|^2\\
&\le 2\eta^2 t\frac{\sigma^2}{n}+2\eta^2 t\left(\frac{1}{n}\sum_{a=0}^{\tau}\sum_{j\in[n]}\left\|\nabla f\left(\mathbf{w}_{k,a}^{(j)}\right)\right\|^2+\right.\\
&\left.\frac{1}{n}\sum_{a=\tau}^{t-1}\sum_{j\in[n]}\left\|\nabla f\left(Q\left(\mathbf{w}_{k,a}^{(j)}\right)\right)\right\|^2\right)+2\eta^2 t\sigma^2\\
&+2\eta^2 t\left(\sum_{a=0}^{\tau}\left\|\nabla f\left(\mathbf{w}_{k,a}^{(i)}\right)\right\|^2+\sum_{a=\tau}^{t-1}\left\|\nabla f\left(Q\left(\mathbf{w}_{k,a}^{(i)}\right)\right)\right\|^2\right).
\end{aligned} \tag{47}$$

For all $i\in[n]$, and summing over the local update round $t=0,1,\cdots,\tau+M-1$, we have

$$\begin{aligned}
&\sum_{t=0}^{M+\tau-1}\sum_{i\in[n]}\mathbb{E}\left\|\bar{\mathbf{w}}_{k,t}-\mathbf{w}_{k,t}^{(i)}\right\|^2\\
&\le\eta^2\sigma^2(n+1)(M+\tau)(M+\tau-1)\\
&+2\eta^2(M+\tau)(M+\tau-1)\\
&\times\sum_{i\in[n]}\left(\sum_{t=0}^{\tau-1}\left\|\nabla f\left(\mathbf{w}_{k,t}^{(i)}\right)\right\|^2+\sum_{t=\tau}^{M+\tau-2}\left\|\nabla f\left(Q\left(\mathbf{w}_{k,t}^{(i)}\right)\right)\right\|^2\right).
\end{aligned} \tag{48}$$

Moreover according to the condition 3 in Assumption 1 being valid and the analyses above, we can get the result as follows:

$$\begin{aligned}
&\mathbb{E}f(\bar{\mathbf{w}}_{k,\tau+M})\\
&\le\mathbb{E}f\left(\mathbf{w}_k\right)-\frac{1}{2}\eta\sum_{t=0}^{\tau-1}\mathbb{E}\left\|\nabla f\left(\bar{\mathbf{w}}_{k,t}\right)\right\|^2\\
&-\frac{1}{2}\eta\sum_{m=0}^{M-1}\left\|\nabla f\left(\bar{\mathbf{w}}_{k,\tau+m}\right)\right\|^2\\
&-\eta\left(\frac{1}{2n}-\frac{1}{2n}L\eta-\frac{1}{n}L^2(M+\tau)(M+\tau-1)\eta^2\right)\\
&\times\sum_{t=0}^{\tau-1}\sum_{i\in[n]}\mathbb{E}\left\|\nabla f\left(\mathbf{w}_{k,t}^{(i)}\right)\right\|^2\\
&-\eta\left(\frac{1}{2n}-\frac{1}{2n}L\eta-\frac{1}{n}L^2(M+\tau)(M+\tau-1))\eta^2\right)\\
&\times\sum_{m=0}^{M-1}\sum_{i\in[n]}\mathbb{E}\left\|\nabla f\left(Q\left(\mathbf{w}_{k,\tau+m}^{(i)}\right)\right)\right\|^2\\
&+\eta^2\frac{L}{2}\frac{\sigma^2}{n}(M+\tau)\\
&+\eta^3\frac{\sigma^2}{n}(n+1)(M+\tau)(M+\tau-1)L^2.
\end{aligned} \tag{49}$$

Lemma 3: Assuming that he condition 1 and 3 in Assumption 1 hold, we have

$$\begin{aligned}
&\mathbb{E}\left\|\hat{\mathbf{w}}_{k+1}-\bar{\mathbf{w}}_{k,\tau+M}\right\|^2\\
&\le q\frac{\sigma^2}{n}(\tau+M)\eta^2+q\frac{1}{n^2}(M+\tau)\eta^2\left(\sum_{t=0}^{\tau-1}\left\|\nabla f_i(\mathbf{w}_{k,t}^{(i)})\right\|^2\right.\\
&\left.+\sum_{m=0}^{M-1}\left\|\nabla f_i\left(Q(\mathbf{w}_{k,\tau+m}^{(i)})\right)\right\|^2\right).
\end{aligned} \tag{50}$$

Proof: According to the condition 1 in Assumption 1

$$\mathbb{E}\left\|\hat{\mathbf{w}}_{k+1}-\bar{\mathbf{w}}_{k,\tau+M}\right\|^2\le\frac{1}{n^2}\sum_{i\in[n]}q\mathbb{E}\left\|\mathbf{w}_{k,\tau+M}^{(i)}-\mathbf{w}_k\right\|^2. \tag{51}$$

According to the local updates in QAT-FL method outlined in Algorithm 1

$$\begin{aligned}
&\mathbb{E}\left\|\mathbf{w}_{k,\tau+M}^{(i)}-\mathbf{w}_k\right\|^2\\
&=\eta^2\mathbb{E}\left\|\tilde{\nabla}f_i(\mathbf{w}_k)+\tilde{\nabla}f_i(\mathbf{w}_{k,1}^{(i)})+\cdots+\tilde{\nabla}f_i(\mathbf{w}_{k,\tau-1}^{(i)})\right.\\
&+\tilde{\nabla}f_i\left(Q(\mathbf{w}_{k,\tau}^{(i)})\right)+\tilde{\nabla}f_i\left(Q(\mathbf{w}_{k,\tau+1}^{(i)})\right)\\
&\left.+\cdots+\tilde{\nabla}f_i\left(Q(\mathbf{w}_{k,\tau+M-1}^{(i)})\right)\right\|^2\\
&=\eta^2\mathbb{E}\left\|\tilde{\nabla}f_i(\mathbf{w}_k)-\nabla f_i(\mathbf{w}_k)+\cdots+\tilde{\nabla}f_i(\mathbf{w}_{k,\tau-1}^{(i)})\right.\\
&-\nabla f_i(\mathbf{w}_{k,\tau-1}^{(i)})+\tilde{\nabla}f_i\left(Q(\mathbf{w}_{k,\tau}^{(i)})\right)-\nabla f_i\left(Q(\mathbf{w}_{k,\tau}^{(i)})\right)\\
&+\cdots+\tilde{\nabla}f_i\left(Q(\mathbf{w}_{k,\tau+M-1}^{(i)})\right)-\nabla f_i\left(Q(\mathbf{w}_{k,\tau+M-1}^{(i)})\right)\Big\|^2\\
&+\eta^2\left\|\nabla f_i(\mathbf{w}_k)+\cdots+\nabla f_i(\mathbf{w}_{k,\tau-1}^{(i)})+\nabla f_i\left(Q(\mathbf{w}_{k,\tau}^{(i)})\right)\right.\\
&\left.+\cdots+\nabla f_i\left(Q(\mathbf{w}_{k,\tau+M-1}^{(i)})\right)\right\|^2\\
&\le\eta^2\sigma^2(\tau+M)+\eta^2(M+\tau)\left(\sum_{t=0}^{\tau-1}\left\|\nabla f_i(\mathbf{w}_{k,t}^{(i)})\right\|^2\right.\\
&\left.+\sum_{m=0}^{M-1}\left\|\nabla f_i\left(Q(\mathbf{w}_{k,\tau+m}^{(i)})\right)\right\|^2\right).
\end{aligned} \tag{52}$$

For all $i\in[n]$, summing up, we can get Lemma 2.

Lemma 4: Selecting r nodes to update from n distributed nodes, we have

$$\begin{aligned}
&\left\|\hat{\mathbf{w}}_{k+1}-\mathbf{w}_{k+1}\right\|^2\\
&\le\frac{1}{r(n-1)}\left(1-\frac{r}{n}\right)4(1+q)\sum_{i\in[n]}\left\|\mathbf{w}_{k,\tau+M}^{(i)}-\mathbf{w}_k\right\|^2\\
&\le\frac{1}{r(n-1)}\left(1-\frac{r}{n}\right)4(1+q)\left\{n\sigma^2(\tau+M)\eta^2\right.\\
&+(\tau+M)\eta^2\times\sum_{i\in[n]}\left(\sum_{t=0}^{\tau-1}\left\|\nabla f_i(\mathbf{w}_{k,t}^{(i)})\right\|^2\right.\\
&\left.\left.+\sum_{m=0}^{M-1}\left\|\nabla f_i\left(Q(\mathbf{w}_{k,\tau+m}^{(i)})\right)\right\|^2\right)\right\}.
\end{aligned} \tag{53}$$

Proof: We definite $\mathbf{z}_{k,\tau+M}^{(i)}=Q\left(\mathbf{w}_{k,\tau+M}^{(i)}-\mathbf{x}_k\right)$,and $\hat{\mathbf{z}}_{k,\tau+M}=\frac{1}{n}\sum_{i\in[n]}\mathbf{z}_{k,\tau+M}^{(i)}$, so we have

$$\mathbb{E}_{S_k}\left\|\mathbf{W}_{k+1}-\hat{\mathbf{W}}_{k+1}\right\|^2$$

$$
\begin{aligned}
&= \mathbb{E}_{S_k}\left\|\frac{1}{r}\sum_{i\in S_k}\mathbf{z}_{k,\tau+M}^{(i)}-\hat{\mathbf{z}}_{k,\tau+M}\right\|^2\\
&= \frac{1}{r^2}\mathbb{E}_{S_k}\left\|\sum\mathbb{I}_{i\in[n]}\{i\in S_k\}\left(\mathbf{z}_{k,\tau+M}^{(i)}-\hat{\mathbf{z}}_{k,\tau+M}\right)\right\|^2\\
&= \frac{1}{r^2}\left\{\sum_{i\in[n]}\Pr\left[i\in S_k\right]\left\|\mathbf{z}_{k,\tau+M}^{(i)}-\hat{\mathbf{z}}_{k,\tau+M}\right\|^2\right.\\
&\quad+\sum_{i\neq j}\left(\Pr\left[i,j\in S_k\right]\right.\\
&\quad\left.\left.\times\left\langle\mathbf{z}_{k,\tau+M}^{(i)}-\hat{\mathbf{z}}_{k,\tau+M},\mathbf{z}_{k,\tau+M}^{(j)}-\hat{\mathbf{z}}_{k,\tau+M}\right\rangle\right)\right\}c\\
&= \frac{1}{nr}\sum_{i\in[n]}\left\|\mathbf{z}_{k,\tau+M}^{(i)}-\hat{\mathbf{z}}_{k,\tau+M}\right\|^2\\
&\quad+\frac{r-1}{rn\left(n-1\right)}\sum_{i\neq j}\left\langle\mathbf{z}_{k,\tau+M}^{(i)}-\hat{\mathbf{z}}_{k,\tau+M},\mathbf{z}_{k,\tau+M}^{(j)}-\hat{\mathbf{z}}_{k,\tau+M}\right\rangle\\
&= \frac{1}{r\left(n-1\right)}\left(1-\frac{r}{n}\right)\sum_{i\in[n]}\left\|\mathbf{z}_{k,\tau+M}^{(i)}-\hat{\mathbf{z}}_{k,\tau+M}\right\|^2. \qquad (54)
\end{aligned}
$$

Further leveraging Assumption 1, we obtain a quantified expectation

$$
\begin{aligned}
&\sum\mathbb{E}_{i\in[n]}\mathbb{E}_Q\left\|\mathbf{z}_{k,\tau+M}^{(i)}-\hat{\mathbf{z}}_{k,\tau+M}\right\|^2\\
&\leq 2\sum_{i\in[n]}\mathbb{E}_Q\left\|\mathbf{z}_{k,\tau+M}^{(i)}\right\|^2+2n\mathbb{E}_Q\left\|\hat{\mathbf{z}}_{k,\tau+M}\right\|^2\\
&\leq 4\sum_{i\in[n]}\mathbb{E}_Q\left\|\mathbf{z}_{k,\tau+M}^{(i)}\right\|^2\\
&= 4\sum_{i\in[n]}\mathbb{E}_Q\left\|Q\left(\mathbf{w}_{k,\tau+M}^{(i)}-\mathbf{w}_k\right)\right\|^2\\
&\leq 4\left(1+q\right)\sum_{i\in[n]}\left\|\mathbf{w}_{k,\tau+M}^{(i)}-\mathbf{w}_k\right\|^2. \qquad (55)
\end{aligned}
$$

Combining (54) and (55), we can get Lemma 3.

With the aforementioned lemmas established, we can prove the convergence rate of Theorem 1

$$
\begin{aligned}
&\mathbb{E}f\left(\mathbf{w}_{k+1}\right)\\
&\leq \mathbb{E}f\left(\mathbf{w}_k\right)-\frac{1}{2}\eta\sum_{t=0}^{\tau-1}\mathbb{E}\left\|\nabla f\left(\bar{\mathbf{w}}_{k,t}\right)\right\|^2\\
&\quad-\frac{1}{2}\eta\sum_{m=0}^{M-1}\mathbb{E}\left\|\nabla f\left(\bar{\mathbf{w}}_{k,\tau+m}\right)\right\|^2\\
&\quad-\eta\frac{1}{2n}\left(1-L-2L^2(\tau+M)\left(\tau+M-1\right)\eta^2\right)\\
&\quad\times\sum_{i\in[n]}\sum_{t=0}^{\tau-1}\left\|\nabla f_i(\mathbf{w}_{k,t}^{(i)})\right\|^2\\
&\quad-\eta\frac{1}{2n}\left(1-L-2L^2(M+\tau)\left(M+\tau-1\right)\eta^2\right)\\
&\quad\times\left(\sum_{i\in[n]}\sum_{m=0}^{M-1}\left\|\nabla f_i\left(Q(\mathbf{w}_{k,\tau+m}^{(i)})\right)\right\|^2\right)\\
&\quad-\eta\frac{L}{2n}\left(q\frac{1}{n}\left(\tau+M\right)\eta+4\left(\tau+M\right)\eta\frac{n-r}{r\left(n-1\right)}\left(1+q\right)\right)\\
&\quad\times\sum_{i\in[n]}\left(\sum_{t=0}^{\tau-1}\left\|\nabla f_i(\mathbf{w}_{k,t}^{(i)})\right\|^2+\sum_{m=0}^{M-1}\left\|\nabla f_i\left(Q(\mathbf{w}_{k,\tau+m}^{(i)})\right)\right\|^2\right)\\
&\quad+\eta^2\frac{L}{2}\left(1+q\right)\left(\tau+M\right)\left(\frac{n-r}{r\left(n-1\right)}4\sigma^2+\frac{\sigma^2}{n}\right)\\
&\quad+\eta^3\frac{\sigma^2}{n}(n+1)\frac{(\tau+M)(\tau+M-1)}{2}L^2 \qquad (56)
\end{aligned}
$$

For sufficiently η, such that

$$
\begin{aligned}
&1-L\left(1+q\frac{1}{n}\left(\tau+M\right)\eta+4\left(\tau+M\right)\eta\frac{n-r}{r\left(n-1\right)}\left(1+q\right)\right)\\
&\qquad-2L^2(\tau+M)\left(\tau+M-1\right)\eta^2\geq 0 \qquad (57)
\end{aligned}
$$

we have

$$
\begin{aligned}
&\mathbb{E}f\left(\mathbf{w}_{k+1}\right)\\
&\leq \mathbb{E}f\left(\mathbf{w}_k\right)-\frac{1}{2}\eta\sum_{t=0}^{\tau-1}\mathbb{E}\left\|\nabla f\left(\bar{\mathbf{w}}_{k,t}\right)\right\|^2\\
&\quad-\frac{1}{2}\eta\sum_{m=0}^{M-1}\mathbb{E}\left\|\nabla f\left(\bar{\mathbf{w}}_{k,\tau+m}\right)\right\|^2\\
&\quad+\eta^2\frac{L}{2}\left(1+q\right)\left(\tau+M\right)\left(\frac{n-r}{r\left(n-1\right)}4\sigma^2+\frac{\sigma^2}{n}\right)\\
&\quad+\eta^3\frac{\sigma^2}{n}(n+1)\frac{(\tau+M)(\tau+M-1)}{2}L^2 \qquad (58)
\end{aligned}
$$

Summing it over the global aggregation round $k = 0, 1, \cdots, K-1$ and rearranging the terms, then we can proof Theorem 1.

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